

Savitribai Phule Pune University
T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination
Lab Course CS-305-MJ-P
Operating System-I

Duration: 3 Hours

Maximum Marks:35

Q.1 Write the simulation program to implement demand paging and show the page scheduling and total number of page faults according to the LRU (using counter method) page replacement algorithm. Assume the memory of n frames.

Reference String : 3,5,7,2,5,1,2,3,1,3,5,3,1,6,2

[15 Marks]

Q.2 Write a C program that behaves like a shell which displays the command prompt '\$'. It accepts the command, tokenize the command line and execute it by creating the child process. Also implement the additional command 'count' as

- a \$ count c filename: It will display the number of characters in given file
- b \$ count w filename: It will display the number of words in given file
- c \$ count l filename: It will display the number of lines in given file

[15 Marks]

Q.3. Oral/Viva

[05 Marks]